

Common Pitfalls and Solutions for Incorporating Vaccine Effects Into Epidemiological Models

Epidemics 2025

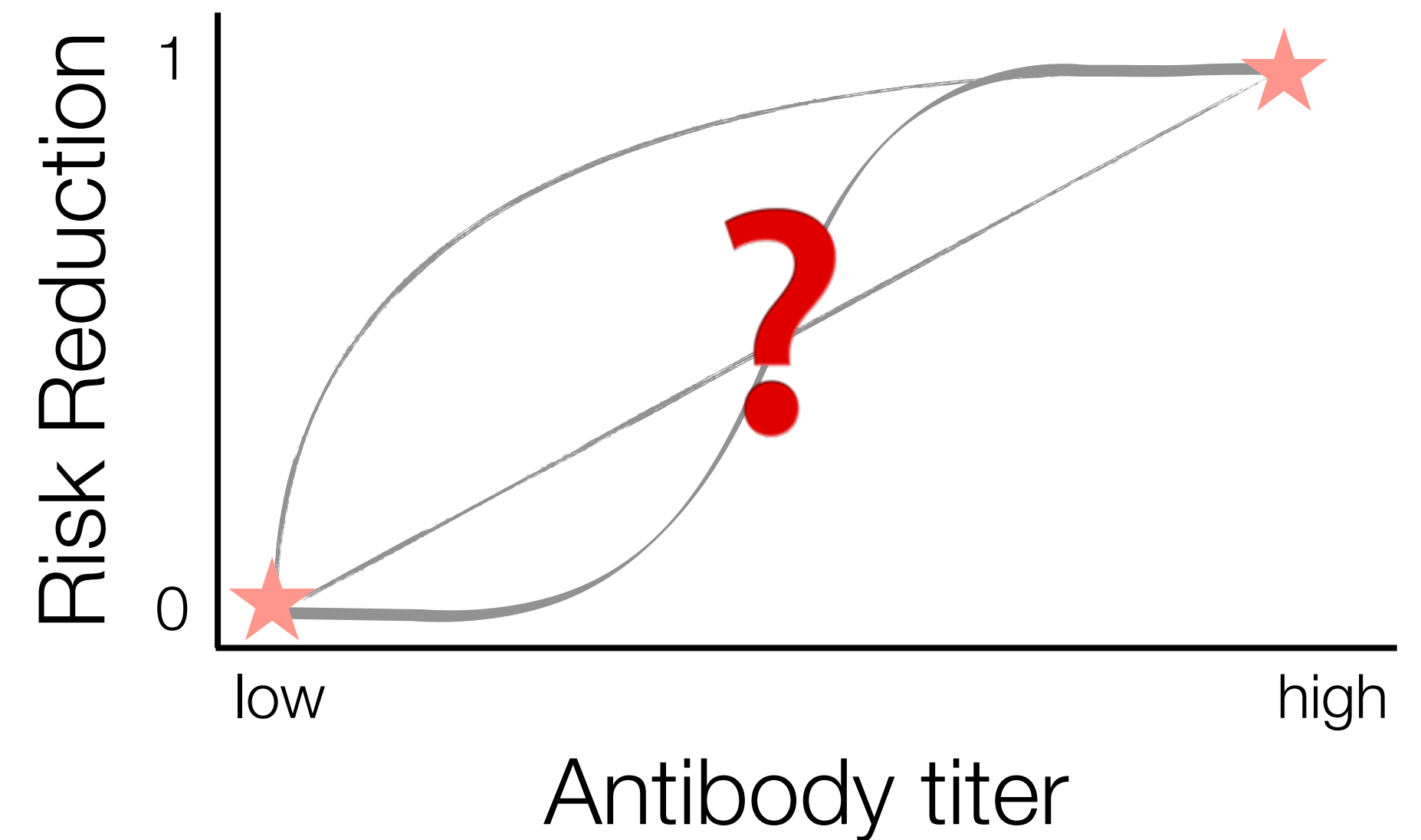
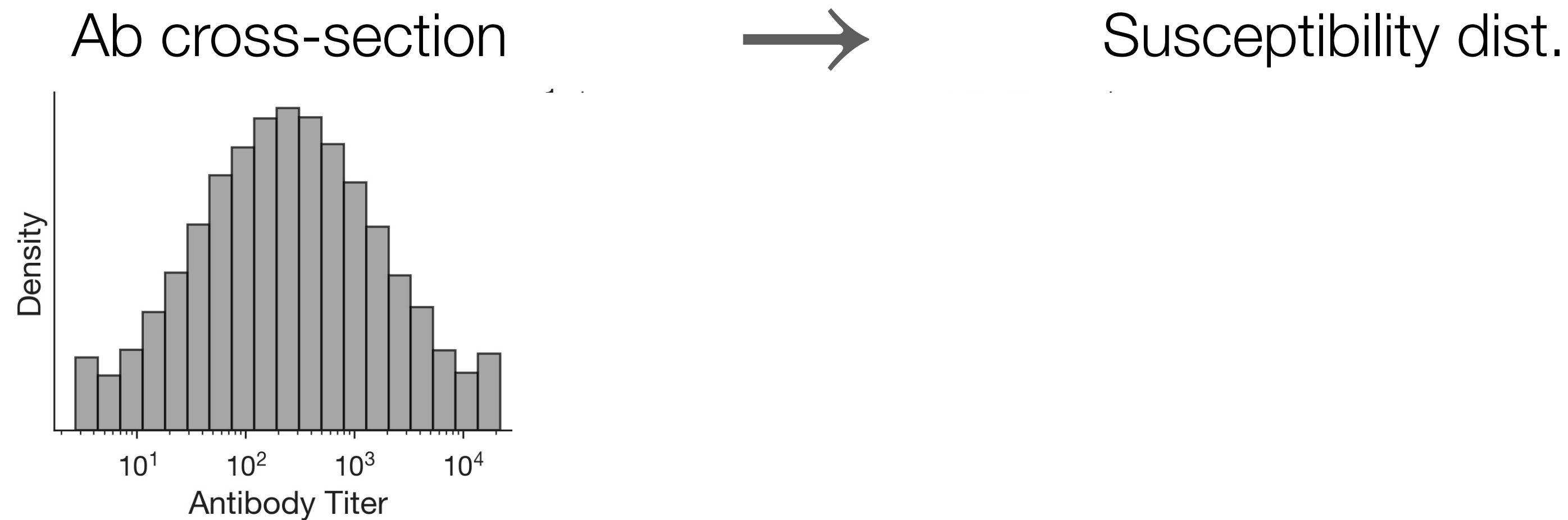
Casey Middleton

Antibodies

Identified as potential **correlates of protection** for

- influenza virus (A. Monto, et al. *JID*, 2015)
- SARS-CoV-2 (E. Nilles, et al. *Lancet ID*, 2013)
- Ebola virus (E. Saphire, et al. *Nat. Immun.*, 2021)
- HIV (J. Mascola, *Vaccine*, 2013)

More antibodies  Lower risk

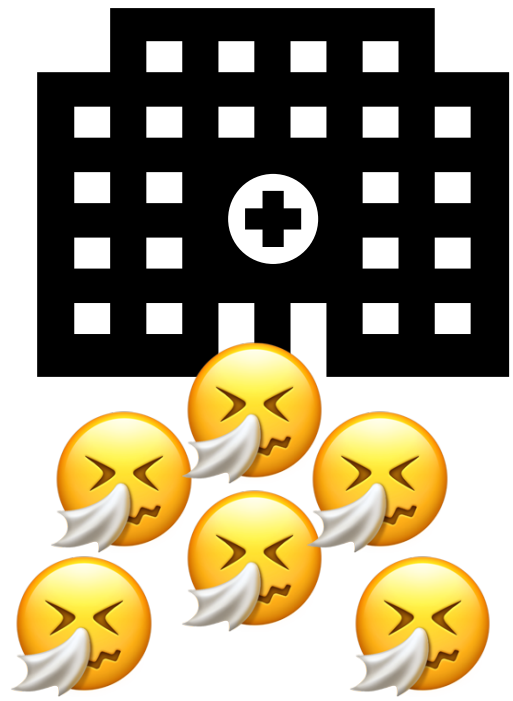


Incorrect protection function → Incorrect downstream analysis

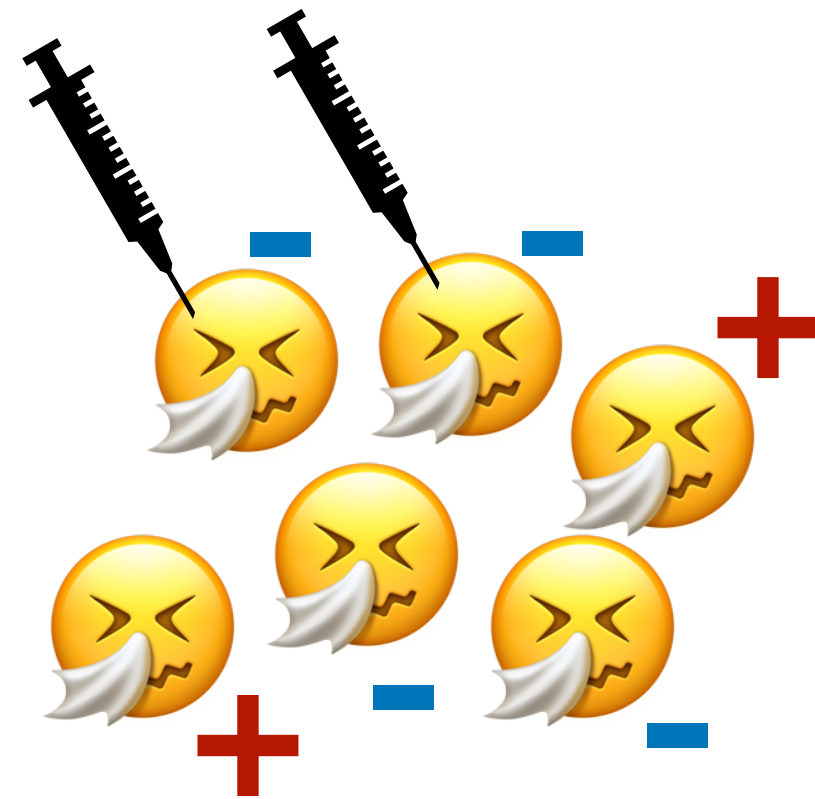
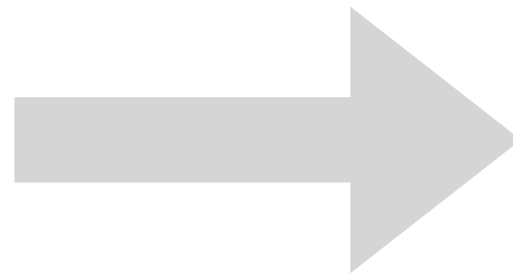
Where do we get data on correlates of protection?

Test-Negative Designs (TNDs)

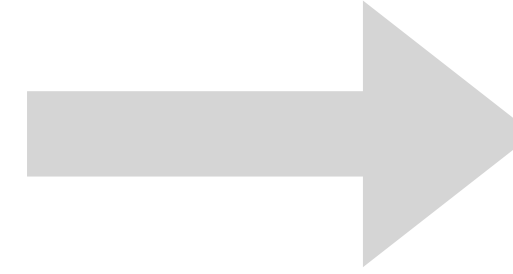
cost-effective study design widely used to estimate vaccine effectiveness (VE)



Recruit patients
with symptoms



Test them for the
focal pathogen



	Vaccinated	Unvaccinated
Infected	6	43
Not Infected	24	96

The **odds ratio** provides an unbiased*
estimator for the *direct effect* of vaccination

Logistic regression is often used to include covariates

$$\text{logit}(p) \sim \beta_0 + \beta_1 V + \beta_\theta \theta$$

V : vaccination status (1/0)

$$VE = 1 - \frac{\text{odds of infection | vaccinated}}{\text{odds of infection | unvaccinated}}$$

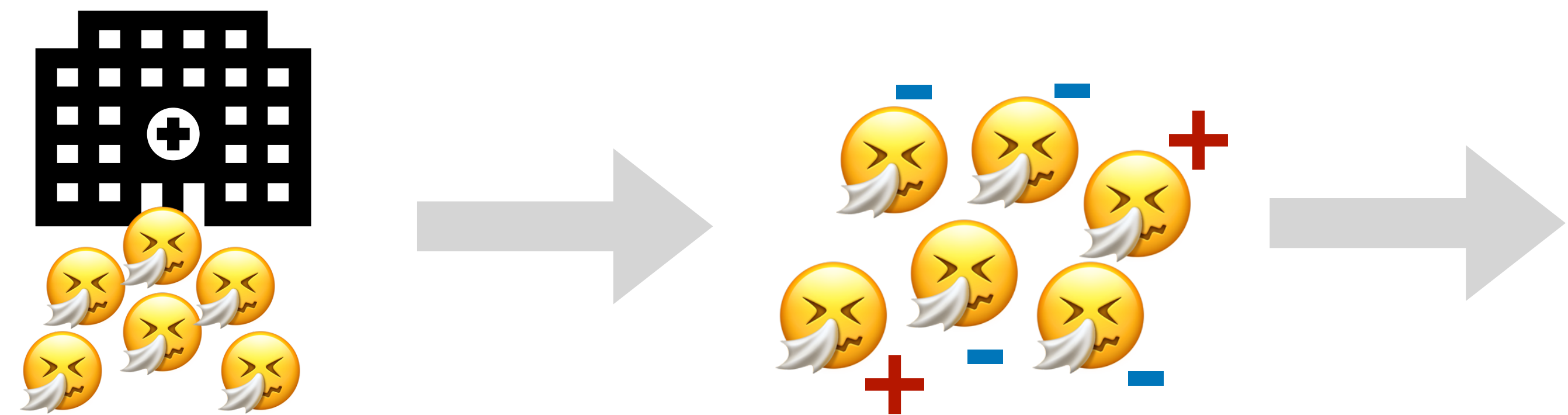
J. Lewnard, et al. *American Journal of Epidemiology*, 2018

*Assuming vaccine decisions are uncorrelated with
exposure, susceptibility, and testing

Test-Negative Designs (TNDs)

More recently, TNDs have been extended to understand **scalar correlates of protection**

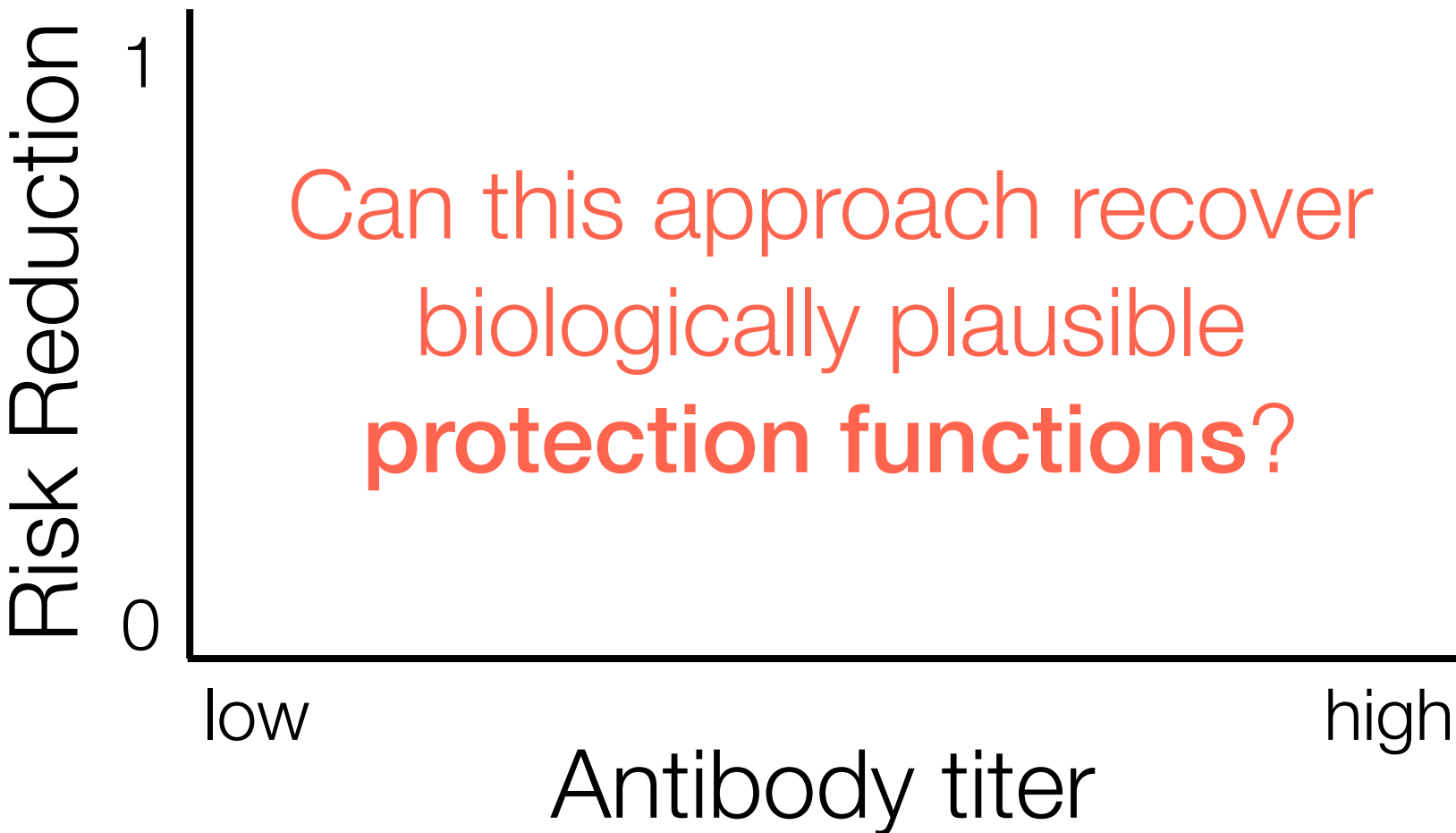
Key assumption:
titer at time of infection
 $\approx$
titer at time of testing



Titer	Infected?
$10^{.39}$	0
$10^{1.22}$	1
$10^{3.28}$	0
$10^{1.43}$	0
$10^{-.11}$	1
$10^{2.09}$	0

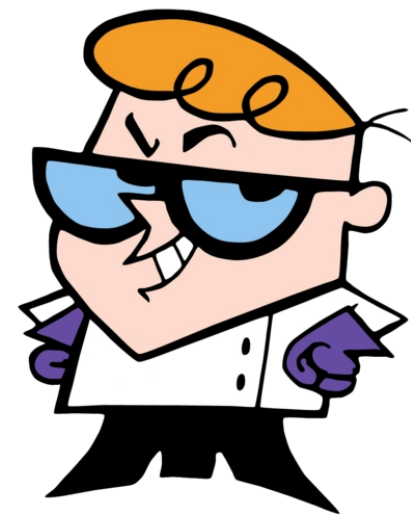
A common approach is using **logistic regression** to estimate odds of infection

Vaccination  Risk reduction
J. Lewnard, et al. *AJE* (2018)



Modelling like an experimentalist

“A first habit is to test whether an analytical pipeline can recover known conditions.”
J. Drake. *Ecology Letters* (2025).



TND Inference Pipeline

- 1 Collect TND data
- 2 Estimate risk using logistic regression
- 3 Compute protection function using odds ratio



Synthetic Data Pipeline

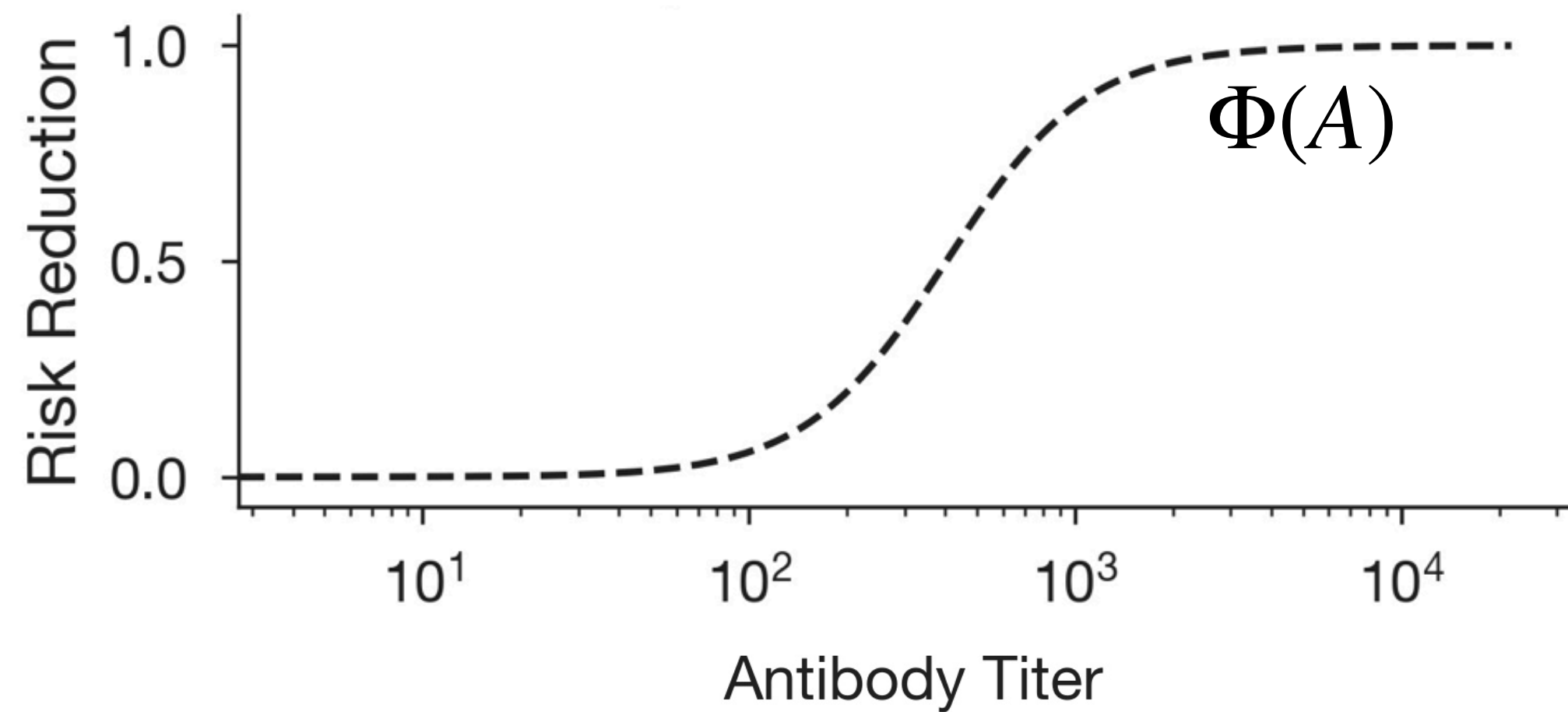


- 1 Define a protection function
- 2 Generate TND data from protection function
- 3 Pass data to TND inference pipeline

Does **3** match **1** ?

Simulating TND data

1 Specify protection function



Case-control studies, animal models, meta-analyses find sigmoidal protection functions.

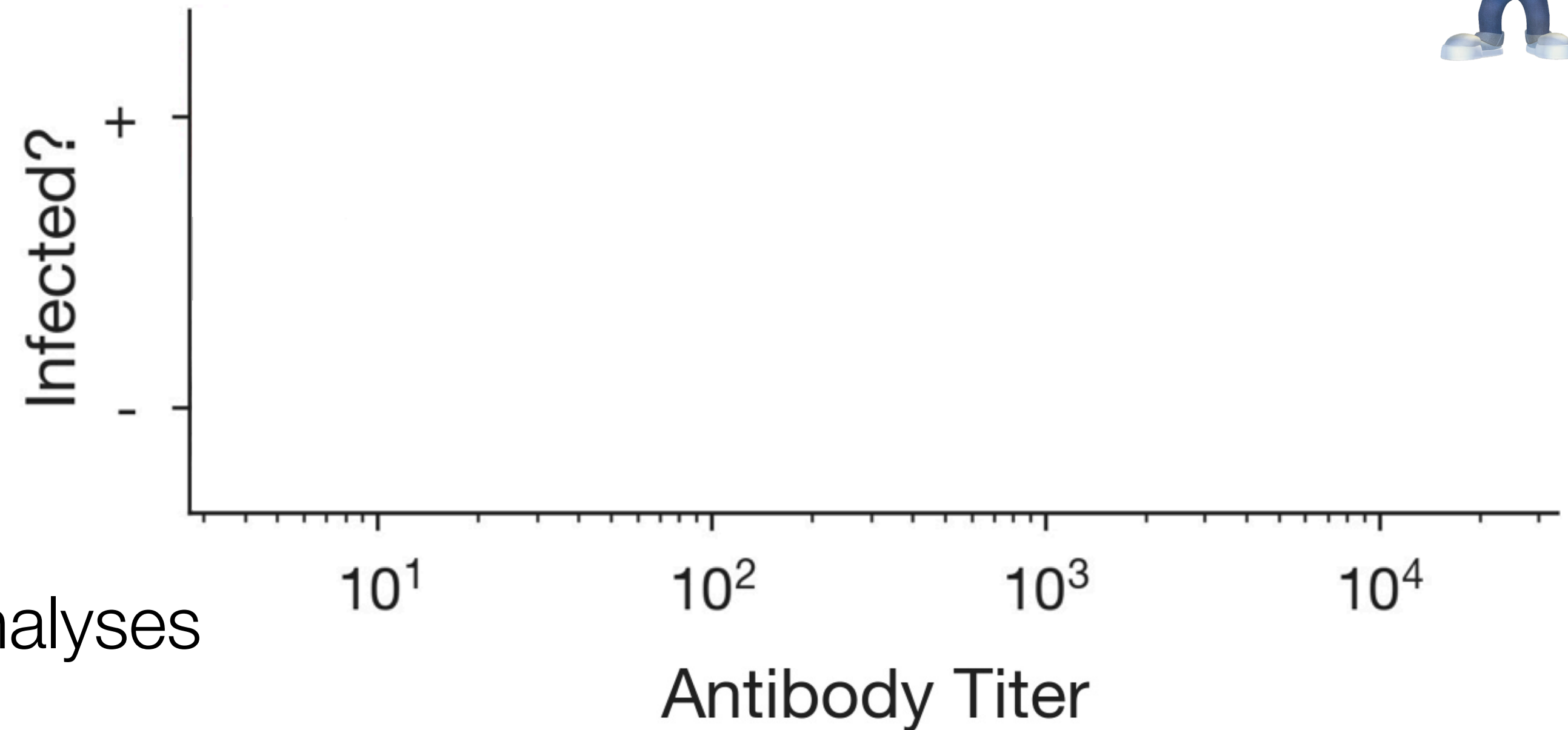
G. Siber, et al. *Vaccine*, 2007

A. Goncalvez, et al. *J. Virol.*, 2008

D. Khoury, et al. *Emerg. Inf. Dis.*, 2023

10^1 10^2 10^3 10^4
Antibody Titer

2 Simulate TND data

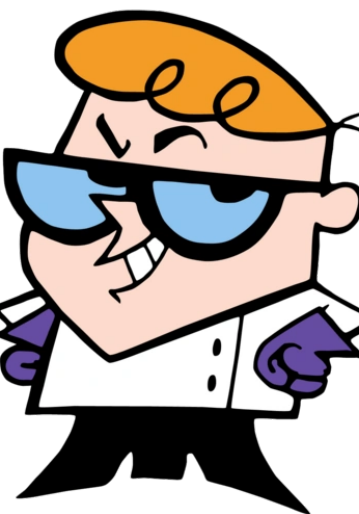
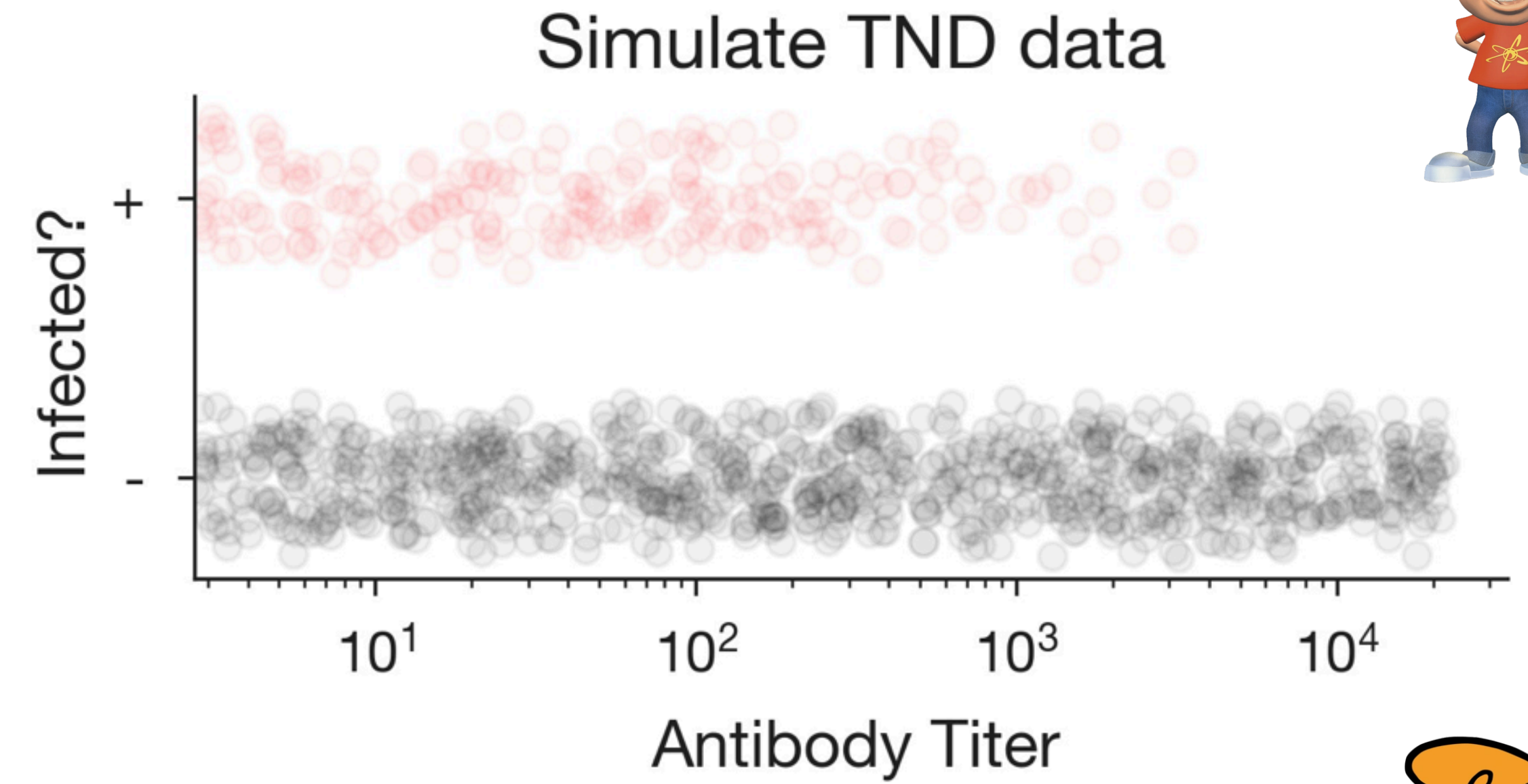
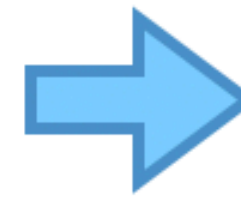
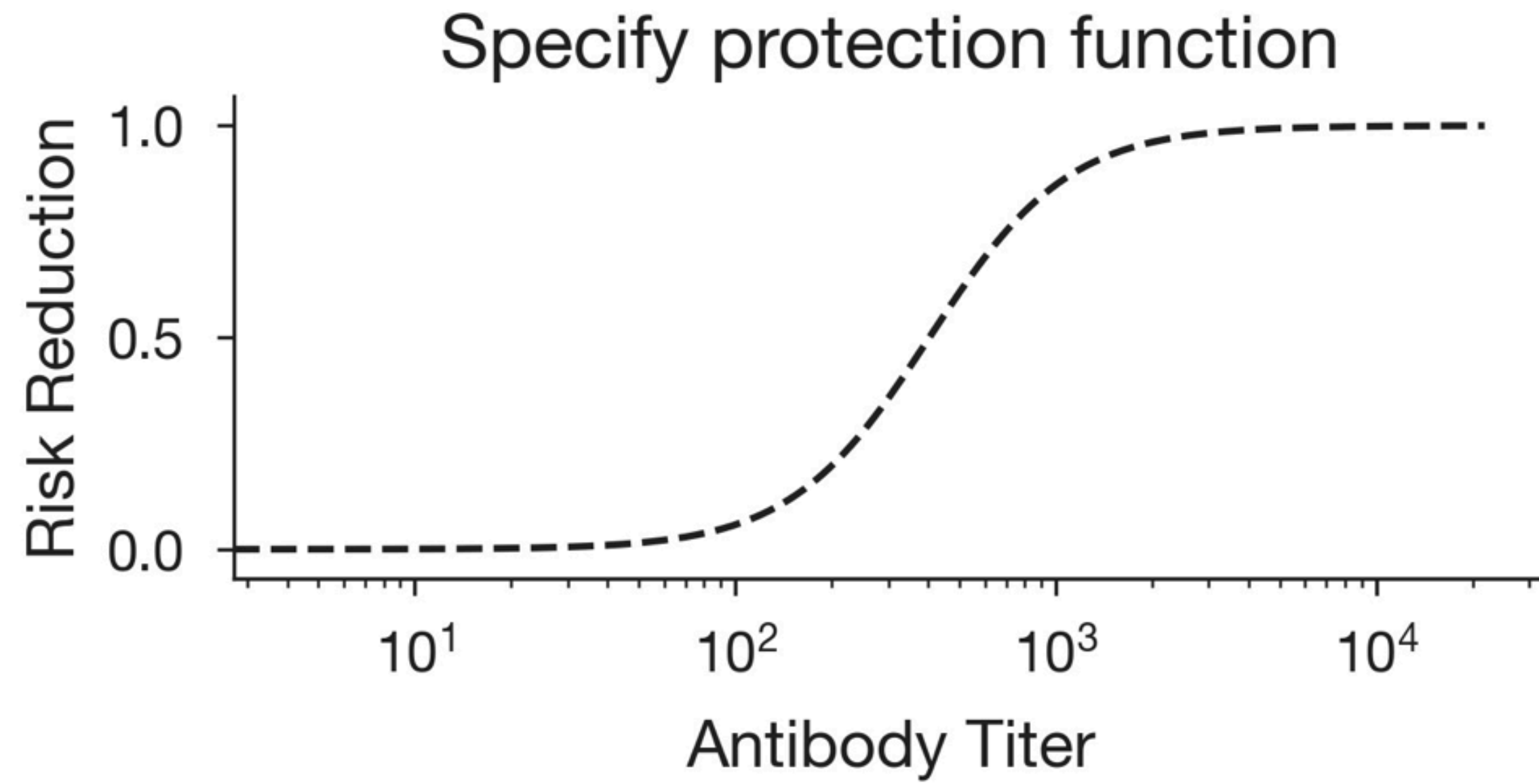


2 Not infected $\sim U(A)$

3 Infected $\sim U(A) \cdot (1 - \Phi(A))$



Results



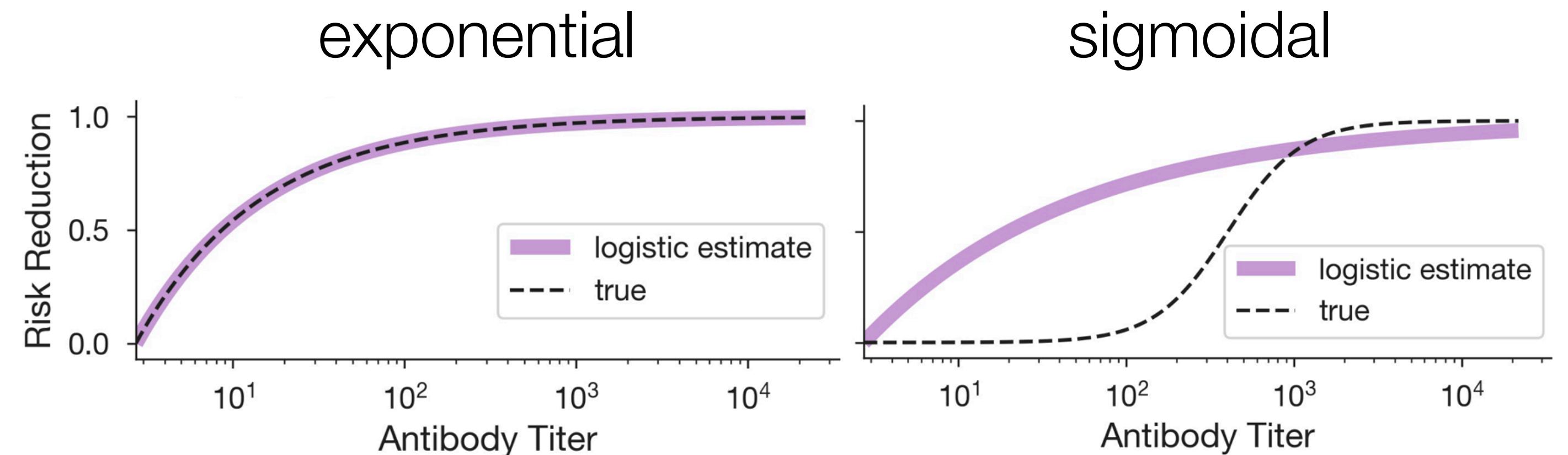
Modeling like an experimentalist: logistic regression

Logistic regression can infer:

✓ exponential

✗ sigmoidal

protection functions from TND data.



This is a result of the odds ratio estimator.

Adding **more terms** does not recover sigmoid

Polynomial logistic regression

$$\text{logit}(p) \sim \beta_0 + \beta_1 A + \beta_2 A^2 + \dots + \beta_n A^n$$

Is there a better alternative model?

Logistic regression is **misspecified** to recover plausible protection functions.
Thus, we need a different model.

Scaled Logit Model

Probability a subject develops disease given by

$$\text{Pr(disease)} = \lambda \cdot (1 - \pi(A))$$

If we assume the protection curve is logistic:

$$\pi(A) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 A)}}$$

Computing the protection function gives

$$\Phi(A) = 1 - OR$$

$$\Phi(A) = 1 - \frac{1 - \lambda + e^{\beta_0}}{1 - \lambda + e^{\beta_0 + \beta_1 A}}$$

λ : probability **susceptible** individual develops disease

$\pi(A)$: probability individual with titer A is protected

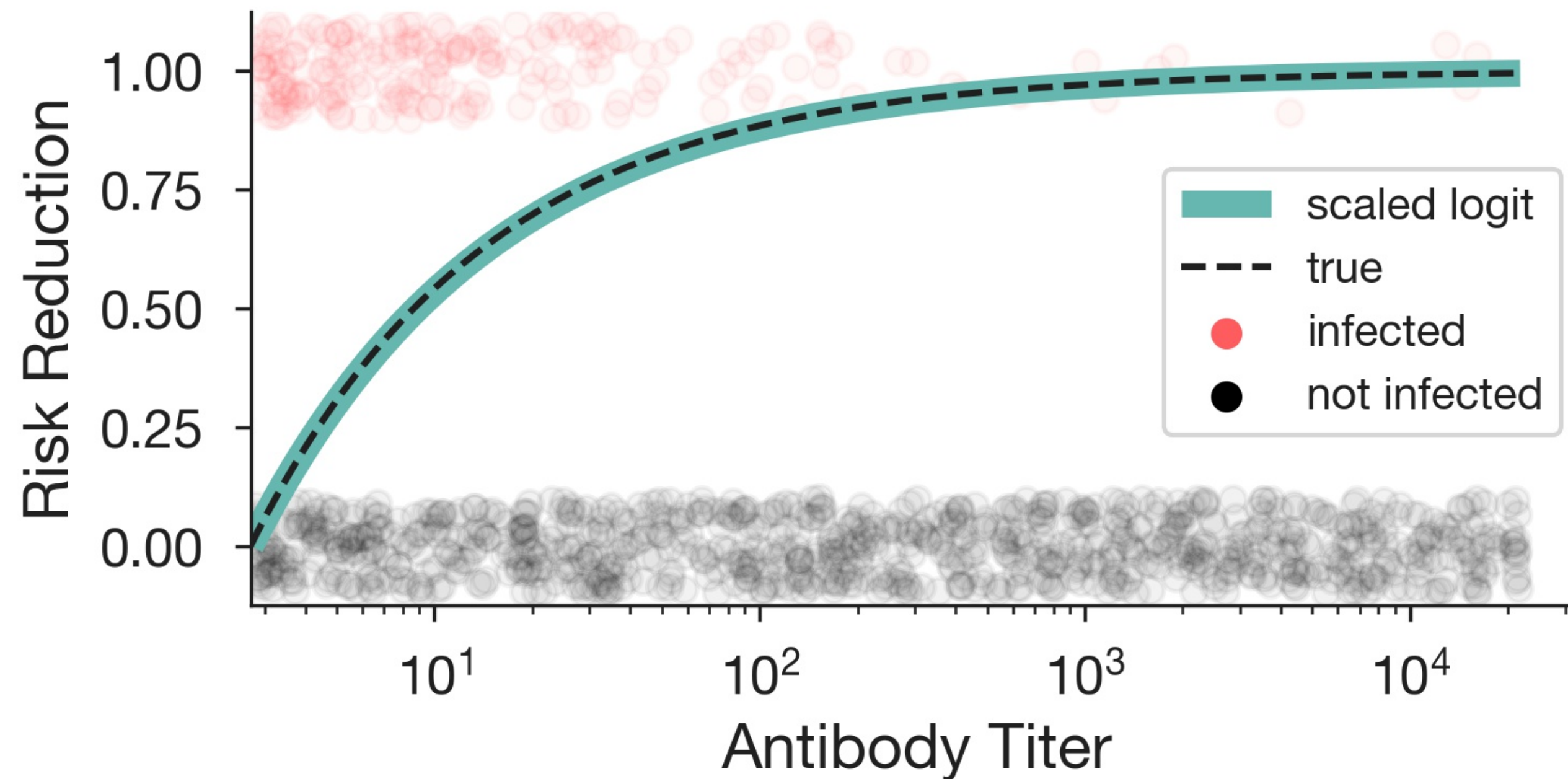
	Logistic	Scaled Logit
Protection Function Estimate	$1 - e^{\beta_1 A}$	$1 - \frac{1 - \lambda + e^{\beta_0}}{1 - \lambda + e^{\beta_0 + \beta_1 A}}$
# fitted parameters	2	3
# parameters in odds ratio (OR)	1	3

Has been used in randomized controlled clinical trials and meta-analyses, but **not in TNDs**.

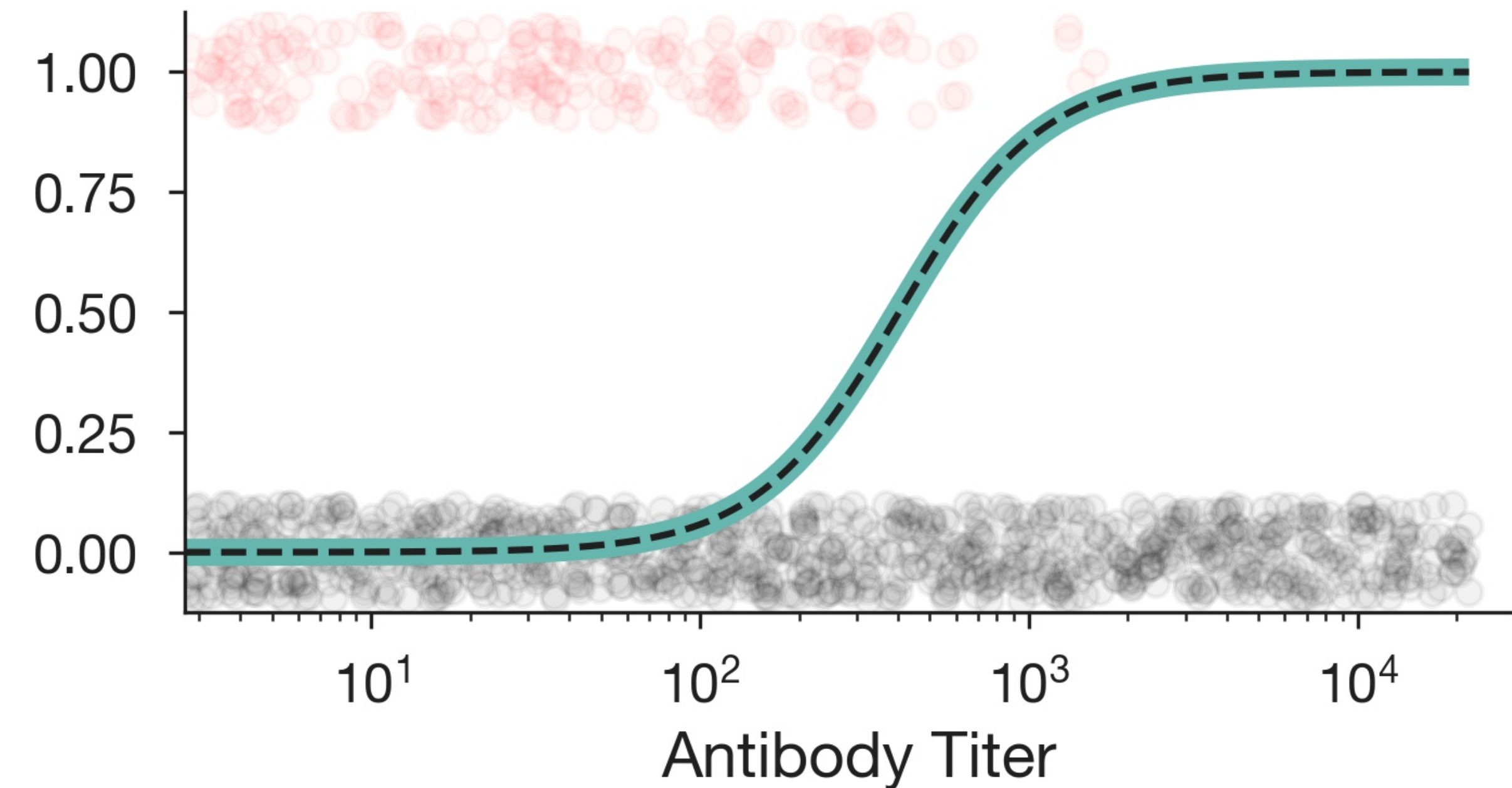
Scaled Logit Model



Exponential



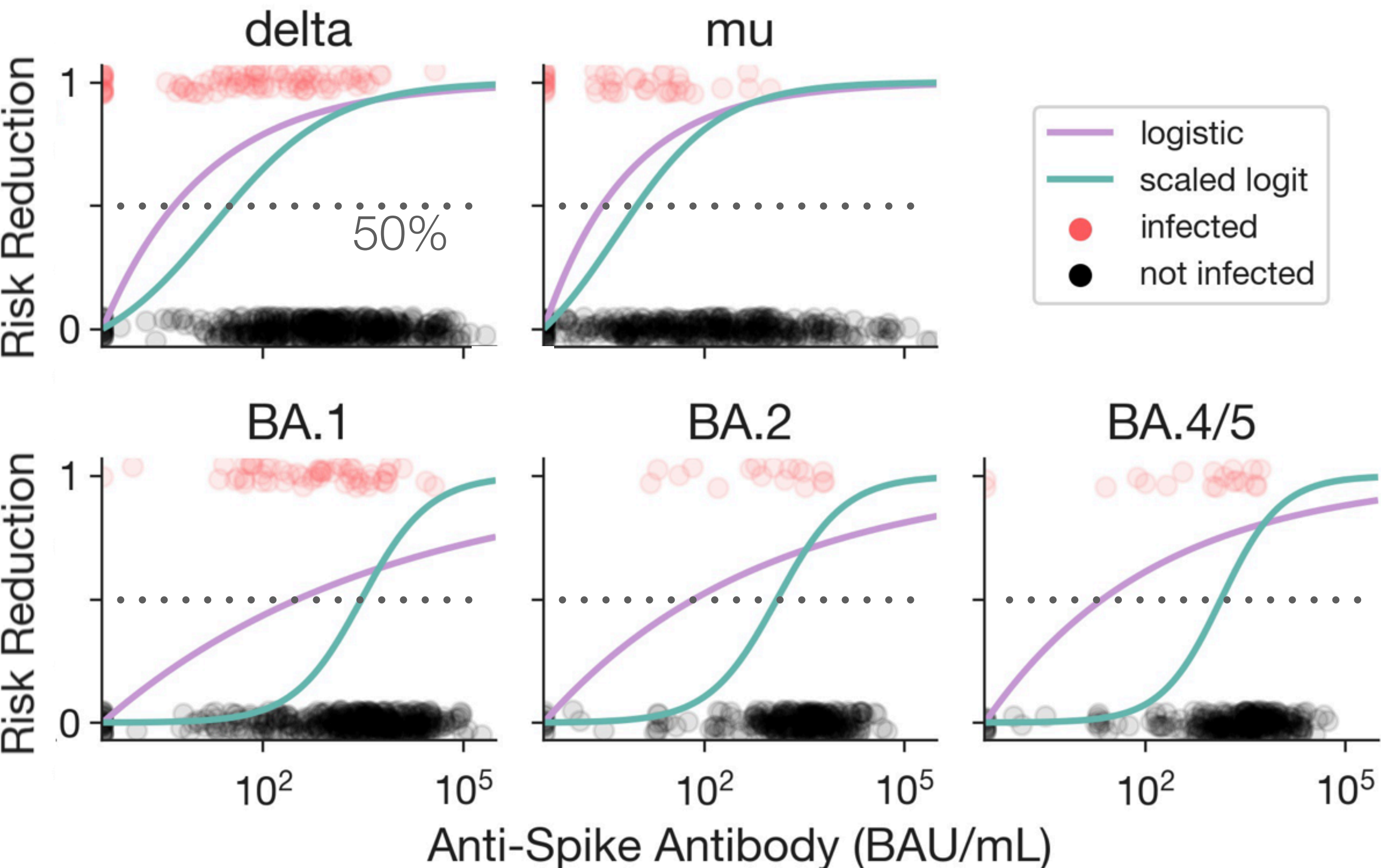
Sigmoidal



The scaled logit model can accurately recover both **exponential** and **sigmoidal** protection functions.

Analysis: How do the logistic and scaled logit models compare when applied to *empirical* data?

Empirical data



Logistic regression always estimates an **exponential**

Scaled logit fit is more **sigmoidal** in shape

AICc		
	Logistic Regression	Scaled Logit
delta	424.3	422.3
mu	247.1	247.9
BA.1	412.1	326.3
BA.2	144.1	129.9
BA.4/5	153.2	139.0

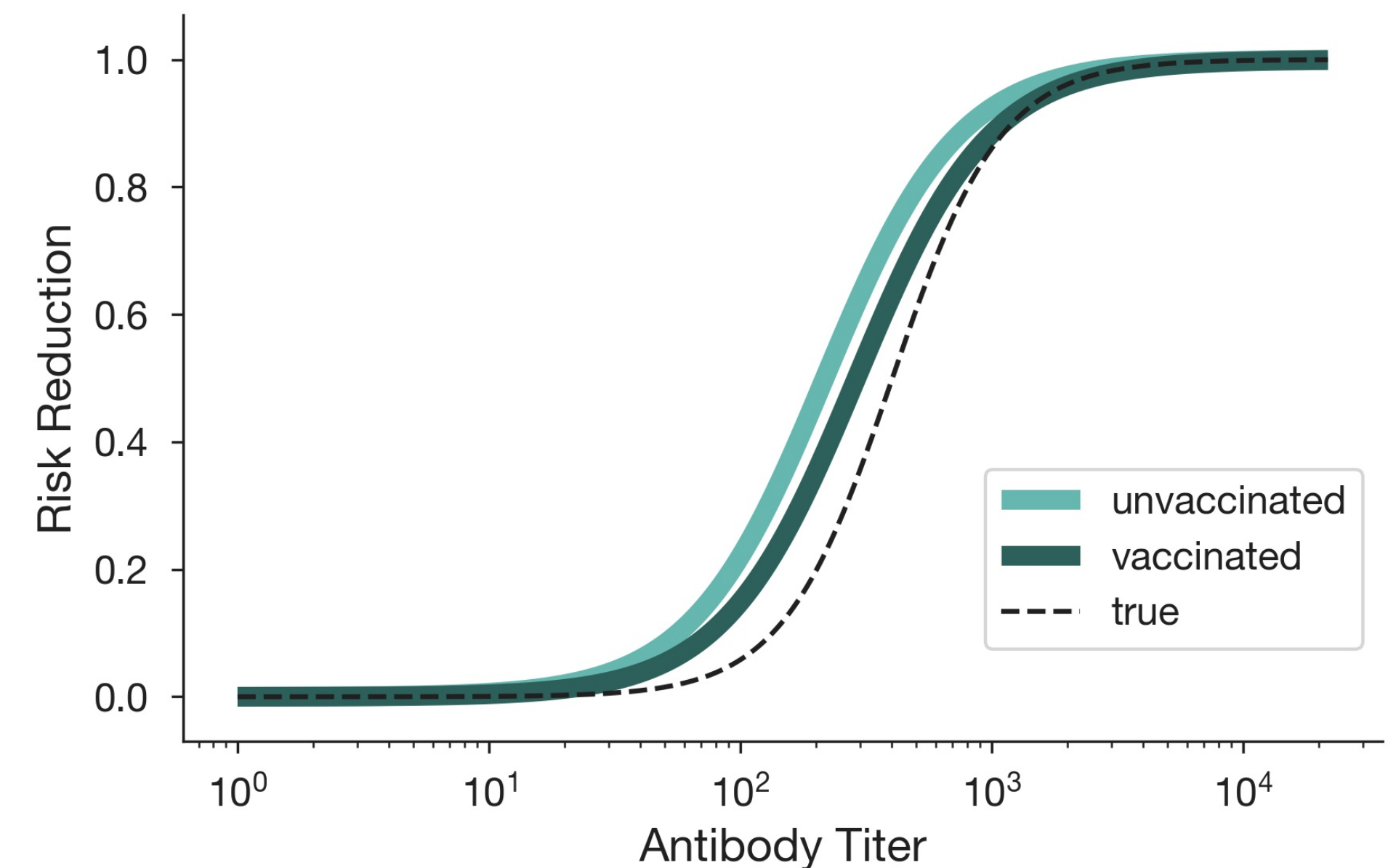
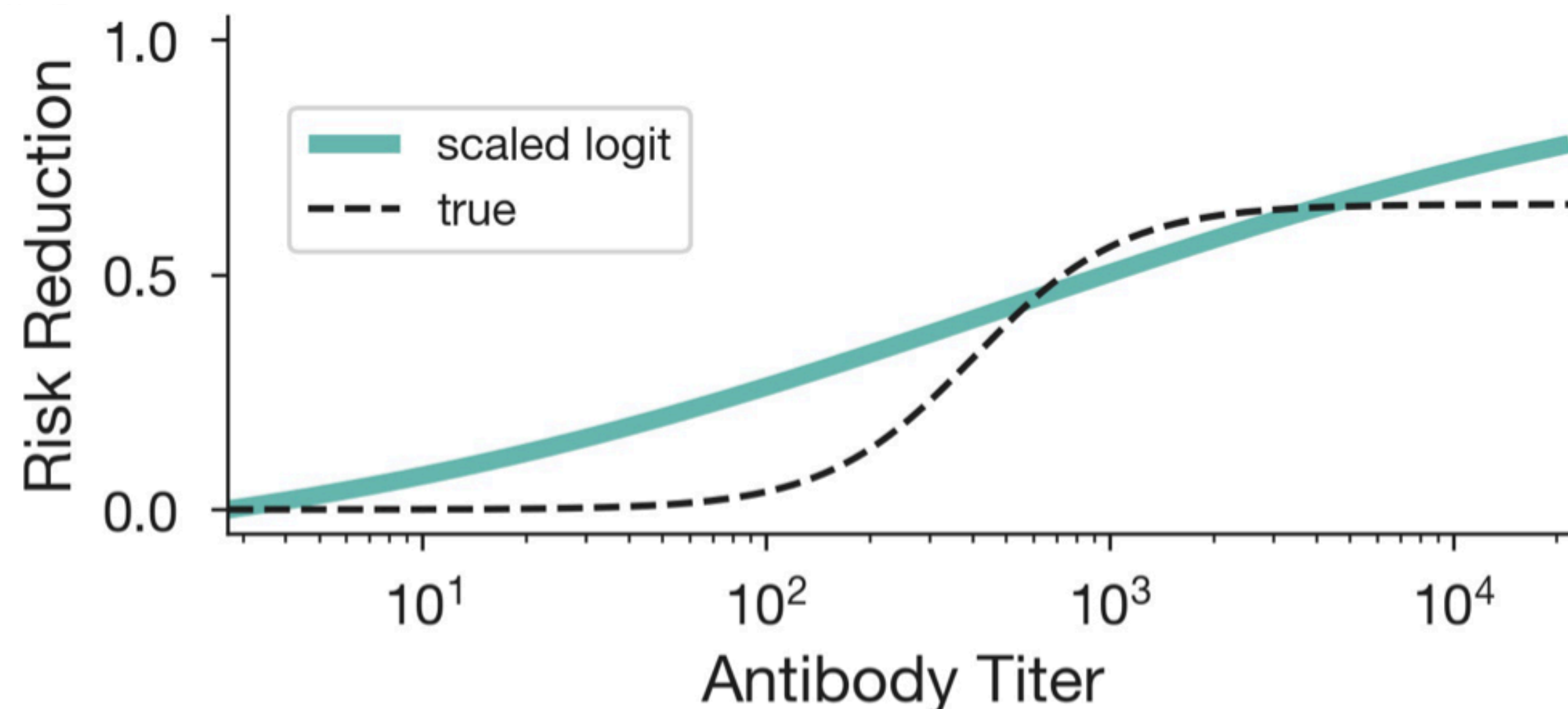
Recall that a scaled logit can accurately infer an exponential or a sigmoid

→ **True curve is likely more sigmoidal**

Limitations

- ❖ Scaled logit model assumes true protection approaches 1 ($\Phi(A) \rightarrow 1$ as $A \rightarrow \infty$)
- ❖ Unobserved covariates that act as colliders (e.g. prior infection) may lead to **underestimation** of titer required to achieve 50% risk reduction
- ❖ Alternative methods:

Generalized additive model (Ziyuan Zhang, Christopher Brian Boyer, and Marc Lipsitch. Use of the test-negative design to estimate the protective effect of a scalar immune measure: A simulation analysis. medRxiv, 2024)



To Review

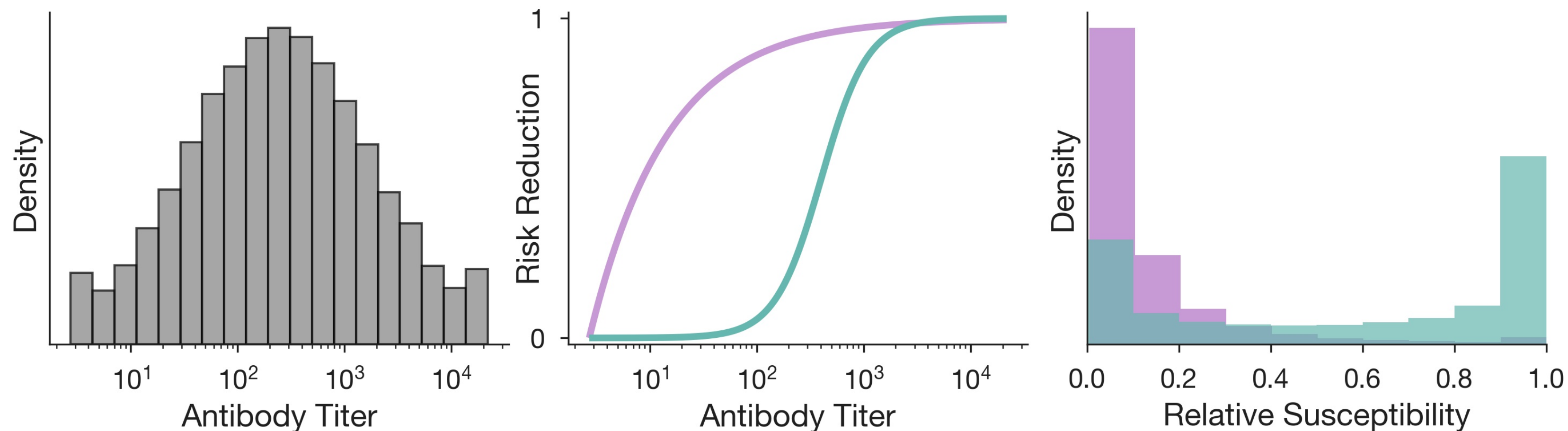
Key results

1. Logistic regression is limited to inferring exponential protection functions in TNDs.
2. Scaled logit model offers more flexibility for inferring protection.

Why model like an experimentalist?

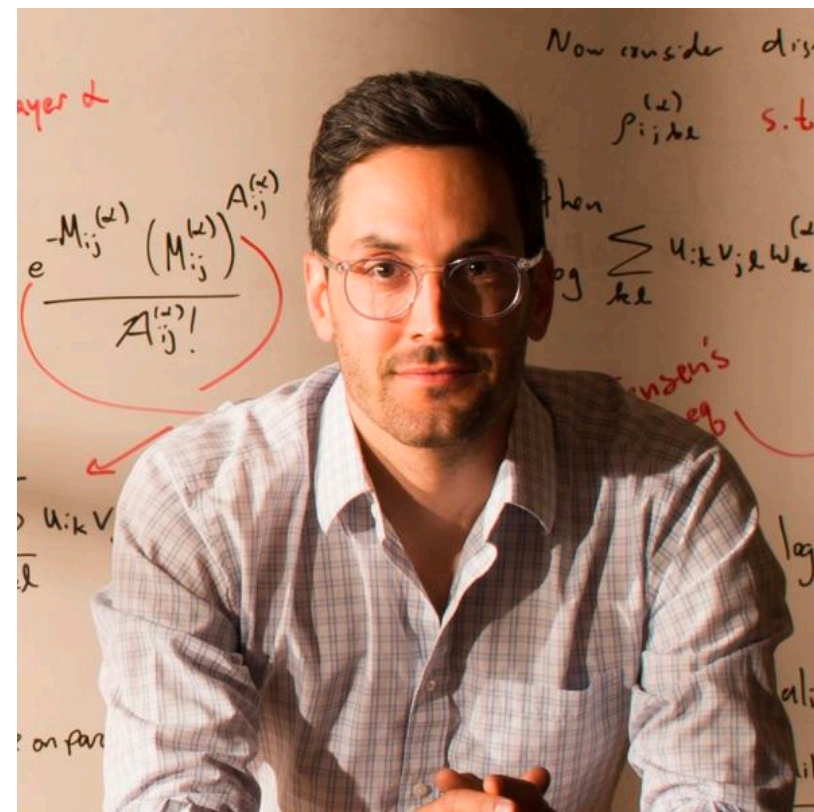
We regularly infer sigmoidal relationships using logistic regression, when the estimator is a risk ratio.

TNDS use an **odds ratio** estimator.



Statistical Methods for Estimating the Protective Effects of Immune Markers in TNDs

Thank you



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C. Middleton and D. Larremore, “Statistical Methods for Estimating the Protective Effects of Immune Markers Using Test-Negative Designs,” *American Journal of Epidemiology* (2025), Forthcoming.